DRILL 02

PPE & SCBA

NFPA 1001 Level 1 (JPR 4.3.1, 4.3.5) / WorkSafeNB Reg 91-191

SCBA EMERGENCY PROCEDURES & AIR MANAGEMENT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Rescue | 🦺 **PPE:** Full Structural Gear & SCBA

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Firefighter Pairs (Buddy System)

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must instinctively handle life-threatening SCBA malfunctions (regulator freeze, runaway free-flow, out-of-air) and execute air conservation techniques (skip breathing). In rural operations with long distances from backup, mastering your air supply is the difference between walking out alive and becoming a Mayday.

📦 TOOLS & APPARATUS SETUP

- 1 SCBA per participant (with working PASS devices and cylinder pressure minimum 2,500 PSI).
- EBSS (Emergency Breathing Safety System / "Buddy Breathing") quick-connect hoses (if equipped).
- Darkened bay or hoods reversed over masks to simulate zero visibility.
- Stopwatch to time skip-breathing intervals.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Hyperventilation Warning:** When practicing skip breathing or air restriction, monitor members closely for lightheadedness or anxiety.

⚠️ **Buddy System (WorkSafeNB 51.6(2)):** Firefighters must stay within arm's reach of their partner during all low-visibility evolutions.

🛑 **THE STOP RULE: Shout "STOP!" immediately if any member experiences real respiratory distress or removes their mask in simulated zero-visibility without safety clearance.**

PHASE 1: TAILGATE WALKTHROUGH & EXPLANATION (10 MINUTES) — SLOW DRY RUN

1. **Regulator Free-Flow / Malfunction:**
   - If the regulator fails open (free-flowing air), open the red **purge/bypass valve** slightly, partially close the **main cylinder valve** to regulate flow, and breathe directly from the flow while exiting immediately.
2. **Skip Breathing / Box Breathing Technique:**
   - Inhale normally -> Hold breath for 5 seconds -> Inhale a little more -> Exhale slowly over 5 seconds.
   - Emphasize: Do not hyperventilate; slow your heart rate down.
3. **PASS Device Activation & Silencing:**
   - Demonstrate manual button press to activate full alarm.
   - Demonstrate how to hold the yellow/reset button to temporarily silence PASS when transmitting a radio Mayday.

PHASE 2: CONTROLLED PRACTICE REPETITIONS (20 MINUTES) — BUDDY COACHING

1. **Malfunction Drill (Full Gear, Lights On):**
   - Firefighter dons SCBA and breathes on air.
   - Leader simulates a stuck regulator. Firefighter:
2. Reaches back to confirm cylinder valve is fully open.
3. Cracks the bypass valve counter-clockwise.
4. Notifies partner by tapping shoulder and giving the "thumbs-out / exit" signal.
5. Transmits simulated radio report: *"Command from Interior, experiencing regulator malfunction, exiting Alpha door."*
6. **Buddy Breathing / EBSS Connection Drill:**
   - Partners kneel facing each other.
   - Firefighter A simulates zero air. Firefighter B unzips/deploys the EBSS buddy hose from their pouch.
   - Firefighter B couples into Firefighter A's intermediate pressure line until it clicks locked.
   - Both confirm steady airflow, grasp each other's harness strap with one hand, and crawl in tandem 20 metres.

PHASE 3: ZERO-VISIBILITY AIR CONSERVATION MAZE (20 MINUTES) — TIMED / STRESS

1. Reverse flash hoods over facepieces (zero visibility).
2. Partners maintain contact (hand on shoulder or 6-foot webbing strap).
3. Crew must crawl through a 30-metre obstacle course (around tires, under tables, over hose) while practicing continuous skip-breathing.
4. Time the evolution: Show firefighters how calming their breathing doubles their remaining air time compared to panic breathing.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Freezing or hesitating to call for help:* If your low-air bell/vibralert sounds, you must already be exiting, not looking for more work.
- ❌ *Forgetting to silence PASS during radio transmissions:* Screaming into the mic over a blaring PASS makes your radio traffic completely unintelligible to the pump operator or IC.
- ❌ *Separating from your partner during buddy breathing:* If you pull apart, the quick-connect can rip or disengage. Keep one hand locked onto your partner's harness.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Who felt claustrophobic or rushed their breathing?", "How easy was it to find your bypass knob with thick gloves on?"***

✅ APPARATUS RESET CHECKLIST

- ☐ All SCBA cylinders swapped and refilled to 4,000–4,500 PSI.
- ☐ EBSS buddy-breathing pouches neatly folded and secured in pouches.
- ☐ PASS devices verified shut off (listen for chirps).
- ☐ Facepieces disinfected and dried.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS